

BIRZEIT UNIVERSITY

COMP322: Human Computer Interaction

Project and Assignments Specification

Second Semester | Dr Hisham Ihshaish

1. Overview

This document specifies the group project, individual assignments, the final presentation, and supporting materials for COMP322. It accompanies the Course Outline and should be read alongside it.

Foundational topics covered in the course so far include: Norman's design principles, the UCD process (ISO 9241-210), usability goals (ISO 9241-11), Nielsen's 10 usability heuristics, the heuristic evaluation method with severity ratings, personas and scenarios, and accessibility foundations (WCAG 2.2 and universal design). These are the tools students will apply in the project.

The project unfolds across three phases that mirror the UCD cycle. Each phase is timed so that the methods required are either already covered by then, or are scaffolded by templates included at the end of this document ([Appendices B and C](#)). In particular, Phase 1 asks teams to conduct a small number of exploratory interviews before formal teaching on data gathering has taken place; the interview guide and consent form templates bridge that gap. Formal techniques for interviewing and usability testing will be consolidated when Block 6 (Data gathering) is taught in early June and Block 8 (Experimental research) follows in mid-June, ahead of the Phase 3 evaluation.

2. Assessment summary

Component	Weight
Multi-phase group project	30%
Final presentation (project defence)	15%
Individual assignment	15%
In-class quiz	10%
Final exam	30%

3. Dates

*Phase submission deadlines are **firm**. Quiz dates and presentation slots will be confirmed in class at least one session in advance.*

Milestone	Date
Teams form	Saturday 18 April onwards
Project brief distributed in class	Wednesday 22 April
Topic proposals due	Monday 27 April
Individual assignment due	Monday 11 May
Phase 1 due	Wednesday 13 May
Quiz	Held during the last week of May
Phase 2 due	Wednesday 3 June
Phase 3 written deliverable due	Wednesday 17 June
Final presentations	Held during final teaching week in June

4. Multi-phase group project (30%)

Project goal

Teams of 3 to 4 students will take a real local digital service or application that has usability problems and redesign a portion of it following the user-centred design cycle. Over the three phases, teams will conduct real user research, produce a working interactive prototype, evaluate it with real participants, and defend their work in a final presentation. Example topics and scope guidance are provided in Appendix A.

Each phase below is presented with its deliverables followed by its marking rubric. Supporting templates (interview guide, consent form, SUS questionnaire) are in Appendices B, C, and D respectively.

PHASE 1 - Research and requirements

Due: Wednesday 13 May, end of day | 30% of project grade

The goal of this phase is to understand who the users are and what they need. Teams use methods already covered in the course (heuristic evaluation, personas, scenarios, usability goals) supplemented by a small number of exploratory interviews supported by the templates in Appendices B and C.

Deliverables (single team-authored report, 5-7 pages plus appendix)

- **Team and topic (half page):** members and working arrangements; chosen topic, scope statement, and justification of why the problem matters.
- **Heuristic evaluation of the existing system (1-2 pages):** evaluation using Nielsen's 10 heuristics; minimum of 10 identified violations, each with the heuristic violated, location (screenshot), description, and Nielsen severity rating (0-4); one paragraph summarising dominant problem patterns.
- **Exploratory user interviews (1-2 pages plus transcripts in appendix):** minimum of 2 interviews per team (3-4 recommended for richer insight), each 20-45 minutes using the interview guide in Appendix B as a starting point; each interviewee must sign the consent form (Appendix C) before beginning; submission includes the guide used, a summary of key findings, and notes from each interview.
- **Personas (1-2 pages):** two personas following the standard template (name, photo, demographics, goals, frustrations, tech literacy, context of use, representative quote), grounded in interview data and observation rather than invented.
- **Scenarios (half to 1 page):** three scenarios describing how the personas would realistically use the redesigned system, written as short narratives.
- **Measurable usability requirements (half page):** 5-7 requirements expressed as measurable targets, mapped to the six ISO 9241-11 usability goals.

Phase 1 rubric

Criterion	Points
Heuristic evaluation is thorough, accurate, and properly documented	25
Interviews are well-conducted (guide used, consent obtained, meaningful summary)	20
Personas are grounded in data and follow proper structure	20
Scenarios are realistic, specific, and connected to personas	15
Requirements are measurable and tied to ISO usability goals	15
Report is clearly written, properly structured, within page limit	5

PHASE 2 - Design and prototype

Due: Wednesday 3 June, end of day | 35% of project grade

The goal of this phase is to produce a design that addresses the requirements from Phase 1. By this point, teams will have covered cognitive aspects (Block 3), social and emotional interaction (Block 4), and the first sessions on interfaces (Block 5). Design decisions must be justified with reference to course concepts, not intuition alone.

Deliverables

- **Design rationale document (3-4 pages):** conceptual model of the redesigned system; how the design addresses each persona's goals and frustrations; specific references to Norman's principles in at least three places; specific references to at least five of Nielsen's heuristics; cognitive load considerations drawing on Block 3; accessibility considerations for keyboard navigation, screen readers, RTL layout for Arabic, and colour contrast (WCAG 2.2).
- **Low-fidelity wireframes:** paper sketches or digital wireframes of at least 8 screens covering all three scenarios from Phase 1; annotations explaining key decisions.
- **Interactive prototype:** built in Figma, Adobe XD, Sketch, or equivalent; functional enough for a user to complete the three scenario tasks; delivered as a shareable link plus PDF export of all screens; includes realistic content and data, not placeholder text.

Phase 2 rubric

Criterion	Points
Conceptual model is coherent and appropriate for the users	15
Design rationale cites Norman's principles with specific examples	20
Design rationale cites Nielsen's heuristics with specific examples	15
Cognitive aspects are addressed (drawing on Block 3)	10
Wireframes cover all scenarios and are adequately annotated	15
Interactive prototype is functional and faithful to the design	15
Accessibility is meaningfully addressed, not an afterthought	10

PHASE 3 - Evaluation and revision

Due: Written deliverable Wed 17 June; defended in final presentation | 35% of project grade

The goal of this phase is to evaluate the prototype against the requirements from Phase 1 and demonstrate the ability to close the UCD loop by iterating. By the time Phase 3 begins, formal teaching on data gathering (Block 6) and experimental methods (Block 8) will have reinforced the techniques used here.

Deliverables

- **Usability test report (3-4 pages plus appendices):** test protocol; at least three test participants distinct from Phase 1 interviewees; think-aloud protocol; signed consent for each participant (Appendix C); quantitative measures (task completion, time on task, error counts); SUS score averaged across participants (questionnaire in Appendix D); qualitative findings organised by theme with anonymised quotes.
- **Heuristic re-evaluation and comparison:** apply Nielsen's 10 heuristics to the new prototype; direct comparison table (original versus redesign); severity ratings for remaining issues.
- **Revision plan (1 page):** prioritised list of what the team would change if given more time, with rationale; demonstrates understanding of iterative design.
- **Reflection (1 page):** what the team learned by going through the UCD process; what was harder or more surprising than expected; how real user research differed from initial assumptions.

Phase 3 rubric

Criterion	Points
Usability test methodology is rigorous and follows think-aloud protocol	20
Quantitative measures are present, accurate, and include SUS	15
Qualitative findings are organised, evidenced, and insightful	20
Heuristic re-evaluation provides genuine before/after comparison	15
Revision plan demonstrates understanding of iterative design	15
Reflection is thoughtful and specific, not generic	15

Project logistics

- Team formation: teams of **3-4 students**, self-organised. Rosters finalised by **Monday 27 April**.
- Topic approval: one-paragraph proposal per team submitted by Monday 27 April.
- Contribution statement: each phase submission must include a signed contribution statement from all team members describing who did what. Uneven contribution affects individual grades within a team.
- Tools: Figma (free tier) is sufficient for all prototyping. Other tools are permitted if accessible to the instructor for review.
- Informed consent: required for every interview and usability test. Use the form in Appendix C. Each participant must sign before the session begins.
- Anonymisation: no identifying information (full names, phone numbers, ID numbers) should appear in any submitted report. Use participant codes (P1, P2, etc.).
- Late submissions: in line with course policy, late submissions are not accepted.

5. Final presentation (15%)

The final presentation is the team's formal defence of Phase 3 work. It consolidates the full UCD cycle into a single narrative.

Format

- 12 minutes of presentation plus 5 minutes of Q&A
- All team members must speak
- Presented during final presentation sessions held across 20-24 June
- Slide deck submitted 24 hours before the team's scheduled slot
- A live demo of the prototype must be included

Expected structure

Segment	Time
Problem and users (Phase 1 summary)	2 minutes
Key requirements	1 minute
Design approach and prototype demo (Phase 2)	4 minutes
Evaluation findings (Phase 3)	3 minutes
What would be changed next (revision plan)	1 minute
Reflection on the UCD process	1 minute
Q&A	5 minutes

Presentation rubric

Criterion	Points
Clarity of problem framing and user research	15
Coherence of design rationale grounded in course concepts	20
Quality of prototype demonstration	20
Rigour of evaluation findings	20
Presentation delivery: clear, well-paced, all members contribute	15
Handling of questions	10

6. Individual assignment (15%)

Heuristic evaluation report

Due: Monday 11 May

Individually, select any public website or mobile application, excluding the team's chosen project topic. Conduct a heuristic evaluation using Nielsen's 10 heuristics. Deliverable: a 2-3 page structured report.

Required content

- Identify a minimum of 8 usability violations
- For each: the heuristic violated, a description with a screenshot, a severity rating (0-4), and a specific design fix
- One paragraph summarising the most severe problem and why it matters

Assignment rubric

Criterion	Points
Heuristics correctly applied where relevant	25
Severity ratings justified by frequency, impact, persistence	20
Design fixes are specific and feasible	25
Screenshots clearly annotated	15
Writing is clear, structured, and within length limit	15

This assignment also serves as individual preparation for the project's Phase 1 heuristic evaluation, which teams conduct collaboratively.

7. In-class quiz (10%)

Held during the last week of May. Exact date confirmed in class one session in advance.

Covers material from the introductory block (UCD, Norman's principles, Nielsen's heuristics, usability goals, heuristic evaluation, personas, accessibility) and Blocks 1-4 (interaction design, conceptualising interaction, cognitive aspects, social and emotional interaction). Format: 20 minutes of short-answer and multiple-choice questions. Material covered after the quiz date (interfaces, data gathering, process, prototyping, experimental methods, data analysis) will be assessed in the final exam and through the project itself.

8. Final exam (30%)

Exam covering all course material. Format: short-answer conceptual questions, scenario-based analysis, and an essay question connecting multiple course themes. Date set by the university.

9. Academic integrity

1. All submitted work must be the team's or individual's own.
2. Proper citation is required for ideas or findings drawn from published sources.
3. AI-generated content may be used for brainstorming or editing support, but analysis, design decisions, and evaluation findings must be the students' own. Substantial use of AI tools must be disclosed in a brief appendix to the submission.
4. Plagiarism and academic dishonesty will be handled according to Birzeit University regulations.

Appendix A: Example project topics

Each team selects one topic from the categories below, or proposes a custom topic with instructor approval. Teams are redesigning a portion of the chosen service, not the entire system. A good scope covers one to three user journeys.

Category A: University services (recommended default)

- (1) Birzeit's Ritaj portal (course registration, grade viewing, schedule management)
- (2) The Birzeit library catalogue or e-resources system
- (3) A campus-wide service such as room booking, lost and found, or event announcements

Category B: Local digital services

- (4) A Palestinian e-government service (e.g. civil affairs, ministry of education)
- (5) A Palestinian bank's mobile application
- (6) A widely used regional service (e.g. a food delivery or ride-hailing application)

Category C: Community needs

- (7) An application supporting commuter students
- (8) A tool for students with disabilities navigating campus
- (9) A digital service for a local NGO or small business, with their consent

Category D: Custom topic

- (10) Teams may propose their own topic in writing (one paragraph). The problem must be real, users must be accessible to the team for interviews and testing, and the scope must be achievable within the project timeline.

Scope guidance: "registering for a course" on Ritaj is a reasonable scope; "the entire Ritaj system" is not. Aim for one to three user journeys that can realistically be researched, designed, and evaluated in eight weeks.

Appendix B: Interview guide template (Phase 1)

This template is a starting point. Teams should adapt it to their specific topic. The goal of an exploratory interview is to understand the participant's experience, not to test them.

Before the interview

- Schedule at a time convenient for the participant (20-45 minutes).
- Conduct in person, via Teams, or by phone, whichever the participant prefers.
- Present and obtain signed consent (Appendix C) before beginning.
- Obtain permission before recording. If not granted, take written notes.
- Have two team members present if possible: one to lead, one to take notes.

Opening script (2-3 minutes)

"Thank you for agreeing to talk with us. We are students in a Human-Computer Interaction course at Birzeit University, and we are studying [your topic in one sentence]. This is not a test of you; we want to learn from your experience. There are no right or wrong answers. You can skip any question you do not want to answer, and we can stop at any time. Do you have any questions before we begin?"

Warm-up questions (3-5 minutes)

Help the participant relax and give context.

- Could you tell us a little about yourself? For a student: year, major, typical day.
- How comfortable are you with technology in general?
- How often do you use [the category of system] in a typical week?

Core questions: current experience (10-20 minutes)

Focus on stories rather than opinions. Stories reveal real behaviour; opinions often reveal what people think they should say.

- Can you walk us through the last time you used [the system]? Start from the beginning: what were you trying to do, where were you, what device were you using?
- When you did [specific task], what happened?
- Was there a moment in that process that frustrated you? Tell us about it.
- Was there a moment that worked well? Tell us about that too.
- What do you wish the system did that it does not do?
- Have you developed any workarounds or tricks to deal with problems?

Interviewing tip: *After each answer, ask follow-up questions such as "Can you say more about that?" or "What happened next?". The richest data comes from follow-ups, not from the first answer.*

Context questions (3-5 minutes)

- Where are you usually when you use this? (On the move? At home? On campus?)
- What device do you usually use? Why that one?
- Is there anyone else involved when you use this (e.g., asking a friend for help)?

Closing (2 minutes)

- Is there anything we did not ask about that you think is important?
- Would you be comfortable if we came back for a follow-up usability test in a few weeks?
- Thank the participant.

After the interview

- Write up notes within 24 hours while memory is fresh.
- Identify 3-5 key themes or quotes from the interview.
- Anonymise: replace names with codes (P1, P2) in any submitted material.
- Compare notes across interviews to spot patterns.

Appendix C: Participant consent form

A consent form must be signed by every interview and usability test participant before the session begins. This template may be used directly or adapted. Teams are encouraged to translate it into Arabic for Arabic-speaking participants; both versions can be offered.

COMP322: Human Computer Interaction

Participant Consent Form

Project title: _____

Team members: _____

Course instructor: Dr Hisham Ihshaish, Department of Computer Science, Birzeit University

What is this study about?

We are students in a Human-Computer Interaction course at Birzeit University. As part of our course project, we are studying how people use [insert system or service name] and looking for ways to improve it.

What will you be asked to do?

You will be asked to (choose one):

- Take part in an interview of approximately 20-45 minutes about your experience with the system.
- Use a prototype of a redesigned system while we observe, for approximately 30 minutes, and answer a short questionnaire afterwards.

We may take written notes, and with your permission, record audio or video. Recordings will only be used by our team for analysis and will be deleted at the end of the course.

What are the risks and benefits?

There are no known risks beyond those of a normal conversation or software use. There is no direct benefit to you, but your input will help us design a better version of the system.

How will your information be used?

- Your identity will not appear in any report or presentation. We will refer to you by a code such as "Participant 1".
- Direct quotes may be used, but only in ways that cannot identify you.
- Data will be stored securely and accessible only to the team and the instructor.
- All raw data will be deleted at the end of the academic semester.

Your rights

- Participation is entirely voluntary.
- You may decline to answer any question.

- You may withdraw at any time without giving a reason and without any consequence. Any data already collected from you will be deleted.

Consent

I have read and understood the information above. I agree to take part in this study.

- ☐ I agree to participate.
- ☐ I agree to be audio or video recorded.
- ☐ I agree to have direct quotes used anonymously in the team's report and presentation.

Participant name: _____

Signature: _____

Date: _____

Team member witness: _____

Appendix D: System Usability Scale (SUS) questionnaire

The SUS is a standard 10-item questionnaire used in usability testing, developed by John Brooke (1986). It produces a single score between 0 and 100 representing overall perceived usability. Use this in Phase 3 after each participant completes the usability test.

Participants respond on a 5-point scale from Strongly Disagree (1) to Strongly Agree (5). For each statement, circle the number that best reflects your view.

#	Statement	SD	D	N	A	SA
1	I think that I would like to use this system frequently.	1	2	3	4	5
2	I found the system unnecessarily complex.	1	2	3	4	5
3	I thought the system was easy to use.	1	2	3	4	5
4	I think that I would need the support of a technical person to be able to use this system.	1	2	3	4	5
5	I found the various functions in this system were well integrated.	1	2	3	4	5
6	I thought there was too much inconsistency in this system.	1	2	3	4	5
7	I would imagine that most people would learn to use this system very quickly.	1	2	3	4	5
8	I found the system very cumbersome to use.	1	2	3	4	5
9	I felt very confident using the system.	1	2	3	4	5
10	I needed to learn a lot of things before I could get going with this system.	1	2	3	4	5

Scoring instructions (for evaluators)

1. For odd-numbered items (1, 3, 5, 7, 9): subtract 1 from the participant's score.
2. For even-numbered items (2, 4, 6, 8, 10): subtract the participant's score from 5.
3. Sum the ten adjusted scores and multiply by 2.5.
4. The result is a score between 0 and 100.
5. Average across all participants for the final SUS score.

Interpretation

Based on Bangor, Kortum and Miller (2008):

Score range	Grade
Below 50	Poor usability
50-70	Acceptable but needs improvement
70-80	Good
80-90	Excellent
Above 90	Best imaginable

Report the final SUS score in the Phase 3 report along with the number of participants and the score's interpretation.

References

- Bangor, A., Kortum, P.T., & Miller, J.T. (2008). An empirical evaluation of the System Usability Scale. *International Journal of Human-Computer Interaction*, 24(6), 574-594.
- Brooke, J. (1986). SUS: A "quick and dirty" usability scale. In P.W. Jordan et al. (Eds.), *Usability Evaluation in Industry*, pp. 189-194. Taylor & Francis.
- ISO 9241-11:2018. Ergonomics of human-system interaction, Part 11: Usability: Definitions and concepts.
- ISO 9241-210:2019. Ergonomics of human-system interaction, Part 210: Human-centred design for interactive systems.
- Nielsen, J. (1994). Enhancing the explanatory power of usability heuristics. *Proc. ACM CHI '94*, 152-158.
- Nielsen, J. (1995). Severity ratings for usability problems. Nielsen Norman Group.
- Rogers, Y., Sharp, H., & Preece, J. (2024). *Interaction Design: Beyond Human-Computer Interaction* (6th ed.). Wiley.
- W3C (2023). *Web Content Accessibility Guidelines (WCAG) 2.2*.